Communication as Governance: Measuring Communicative Stance in University Intellectual Property Policies

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Abstract

University intellectual-property policies allocate rights and also communicate authority. We develop the Policy Communication Stance Index (PCSI), a human-anchored measure of supportive versus restrictive language, using 87,159 sentences from 518 policy documents at 150 U.S. research universities (1944–2025). As scored by our classifier, written stance leans restrictive: 86 percent of university-years fall below the scale midpoint. Stance has become more restrictive since 1980, within universities, policy topics, and rule forms. Universities also differ persistently, and observable characteristics explain little of these differences. The index does not reduce to readability, legal density, or measured rule form.

Keywords: institutional governance; policy communication; university technology transfer; intellectual property; text as data; organizational heterogeneity
JEL Codes: O34, O38, I23, C55, K11

1 Introduction

Universities govern faculty inventions through written intellectual-property (IP) policies. These policies decide who owns an invention, when faculty must disclose it, how rights pass to the university, and how licensing income is shared with inventors. They are the formal backbone of university technology transfer. Most research treats them as sets of rules. But a policy is also a piece of institutional writing, and the way it states its rules may matter in its own right.

The literature on university innovation has studied these rules closely, along with the incentives they create, the offices that administer them, and the commercial activity that follows (Mowery et al., 2001; Sampat, 2006; Jensen et al., 2003; Perkmann et al., 2021). Even so, the link between formal terms and outcomes has been hard to pin down. Lach and Schankerman (2008) reported that universities giving inventors a larger royalty share earn more licensing income; Ouellette and Tutt (2020) later found errors in how those shares were coded and no robust evidence of the effect. What the literature has examined much less is how the policies communicate: whether they address inventors with commands or with offers of help.

We call this *communicative stance*. Two policies can impose the same disclosure duty, one by stating what inventors must do and the other by describing the help the university will give once they come forward. This difference does not require every inventor to read the policy. Faculty usually meet IP governance through technology-transfer offices (TTOs), licensing staff, disclosure procedures, and departmental colleagues, and the written policy supplies the rules these intermediaries interpret and apply (Jensen et al., 2003; Siegel et al., 2003). Studies of faculty behavior show that the perceived costs of dealing with the TTO and the local working environment shape whether faculty disclose inventions and take part in technology transfer (Owen-Smith and Powell, 2001; Bercovitz and Feldman, 2008). Written stance is the first link in that chain, and it is the one we can observe directly.

We measure it with the Policy Communication Stance Index (PCSI). The data cover 150 U.S. research universities from 1944 to 2025: 87,159 cleaned sentences from 518 policy documents. Human coders labeled a sample of sentences as restrictive, neutral, or supportive. A fine-tuned language classifier then extends those judgments to the full corpus, and we average its predicted probabilities into a score for each university and year. This follows the text-as-data practice of defining a concept through human judgment before scaling it statistically (Grimmer and Stewart, 2013; Gentzkow et al., 2019; Adcock and Collier, 2001).

Three findings emerge. First, policy language leans restrictive: mean PCSI is 0.437 on a scale from 0 to 1, and 86 percent of university-years fall below the neutral midpoint. Second, stance has become more restrictive since about 1980. The decline appears within universities and is not confined to the ownership and assignment clauses that a well-known court decision brought into focus. Third, universities differ in lasting ways. About half of the variation lies between universities, and private R1 universities write more supportive policies than public R1 universities, yet observable characteristics explain only a small share of these differences.

The paper's contribution is to make a neglected part of university governance measurable.

Research on technology transfer has asked what IP policies require; we ask how they say it, and we show that the answer varies across institutions and over time in ways that simple features of legal prose do not explain. Section 2 reviews the related literature, Section 3 describes the index, Section 4 presents the findings, and Section 5 discusses what they mean.

2 Related Literature and Conceptual Motivation

2.1 University IP Governance and Innovation Incentives

After the Bayh–Dole Act of 1980, U.S. universities expanded patenting and licensing, and research has followed how this activity grew and what drove it (Mowery et al., 2001; Sampat, 2006). A central concern is incentives. University policies allocate ownership and set the inventor's share of licensing income, so economists have asked whether more generous terms lead to more disclosure and licensing (Lach and Schankerman, 2008; Ouellette and Tutt, 2020). A broader literature on academic engagement places patenting and licensing within a wider set of interactions between academics and external organizations, including collaborative research, contract research, and consulting (Perkmann et al., 2021). Across this work, the policy enters mainly through its terms.

2.2 Organizational Communication and Formal Rules

Organization scholars treat formal texts as more than containers for rules. Texts represent the organization, coordinate activity across settings, and can continue to speak for an institution long after their authors have left (Smith, 2001; Cooren, 2004; Putnam and Cooren, 2004; Fairhurst and Putnam, 2004). Institutional analysis makes a related distinction. The Institutional Grammar breaks a rule into its parts: who is addressed, what is required or allowed, under what conditions, and with what consequence for noncompliance (Crawford and Ostrom, 1995). The same requirement can therefore be stated in different ways, and the wording can signal how the institution expects to relate to the people it governs (Phillips et al., 2004). Work on the expressive function of law makes a similar point: rules send messages beyond their formal sanctions (Sunstein, 1996).

2.3 TTOs as Intermediaries between Policy and Faculty

Written policy can matter for institutional practice whether or not faculty read it directly, because the people and procedures that administer IP governance work from it. Jensen et al. (2003) model the TTO as an agent of both the university administration and the faculty, balancing the university's ownership interest against the need to attract disclosures. Interviews with managers, entrepreneurs, and administrators point to organizational practices, including faculty reward systems, TTO staffing, and efforts to reduce informational and cultural barriers, as important for how well technology transfer works (Siegel et al., 2003). Fieldwork at two universities finds that faculty decisions to disclose depend on the perceived benefits of patenting and the perceived costs of dealing with technology-transfer staff, and on whether the local environment supports

commercial work (Owen-Smith and Powell, 2001). Faculty participation also depends on the department: when the chair and peers are active, others tend to follow (Bercovitz and Feldman, 2008).

These studies suggest a possible pathway from written policy to inventors. The policy states the rules; TTO staff and administrators interpret and apply them; faculty experience governance through those interactions; and that experience may shape their decisions. Our data cover only the first step. The pathway does not assume that TTO staff repeat the wording or tone of the policy, and whether written stance matches actual practice is a question for future research.

2.4 Why Measure Stance Separately from Rule Content

If wording can vary while the rule stays fixed, stance should be measured on its own rather than inferred from a policy's terms. Theory and experiments on controlling incentives give one reason it could matter: signals of control can reduce the effort of people who are intrinsically motivated (Bénabou and Tirole, 2003; Falk and Kosfeld, 2006). The closest existing measure, the survey-based Policy Communication Index of Canary et al. (2013), captures how organizations communicate their policies through practices and channels. We study the stance written into the policy itself.

The literature also suggests what a meaningful measure should show. Stance should vary among sentences with the same grammatical rule form, so that it is not simply a count of obligations. Because formal language is often copied and revised in small steps (Kahan and Klausner, 1997; Choi et al., 2013), stance should persist within each university and be only loosely tied to broad institutional categories. And because organizations in the same field respond to shared legal and professional pressures (DiMaggio and Powell, 1983; Edelman, 1992), any drift in stance should appear across many universities and many kinds of clauses. Section 3.5 examines the first point and Section 4 the other two.

3 Measuring the Policy Communication Stance Index

3.1 Policy Texts

The corpus contains IP policies from 150 U.S. research universities between 1944 and 2025: 87,159 cleaned sentences from 518 policy documents, each identified by university, year, and file. The documents cover ownership, disclosure, assignment, licensing authority, commercialization procedures, and revenue sharing.

When a university has more than one relevant document in a year, we pool their sentences before scoring. This gives 480 university-years with policy text. Between those years, we carry the most recent score forward until a newer document appears, which produces 4,277 annual observations of the policy in force through 2025. Each annual row records the year of the document behind its score, so we can always separate the annual panel from the 480 years with new text. Table 1 summarizes the sample, and Appendix A gives details.

Table 1: Corpus and Analysis Sample, 1944–2025

Quantity	Value
Universities	150
Analysis period	1944–2025
Source policy documents	518
Cleaned sentences	87,159
University-years with source policy text	480
Annual policy-in-force university-years	4,277

Notes: When several source documents are available for the same university and year, their sentences are pooled before the university-year score is constructed. In the annual panel, that score is carried forward until a newer policy source becomes available.

3.2 Human Coding

We define communicative stance as the supportive, neutral, or restrictive way a policy presents rights, obligations, authority, and assistance. Supportive sentences offer help, guidance, encouragement, collaboration, or shared interests. Restrictive sentences stress mandatory assignment, categorical duties, unilateral authority, prohibitions, sanctions, or compliance. Neutral sentences mainly define terms, describe arrangements, or set out procedure. Coders labeled the main message of each full sentence, and disagreements were settled by adjudication. The concept concerns expression, not the generosity of the underlying rule or how readers react.

Rule content and stance need not line up. Table 2 shows examples from the corpus; the last two rows are disclosure provisions written with opposite stance. Appendix B gives the full coding rubric.

Table 2: Illustrative Sentence-Level Communicative Stance

Class	Illustrative sentence	$\hat{p}_{\text{class}}$
Supportive	"Inventors are encouraged to consult with the OTT concerning their options regarding the release of intellectual property."	0.973
Neutral	"Special procedures relevant to these processes are posted on the website of the Office of the Vice President for Research and Innovation."	0.980
Restrictive	"As a condition of affiliation with the University, members of the University community are bound by all University policies, including this one."	0.982
Two disclosure provisions, different framing		
Supportive	"The University intends that the obligation for disclosure shall not unduly interfere with prompt publication of research results."	0.975
Restrictive	"... Each Invention must be disclosed promptly upon its creation, conception or discovery."	0.978

Notes: $\hat{p}_{\text{class}}$ is the calibrated model probability assigned to the displayed class. Source institutions and years are recorded in the replication package. The last two rows both concern invention disclosure but do not impose identical legal terms; they illustrate how provisions serving the same policy function can differ in stance. The examples illustrate the annotation construct; they are not an independent accuracy test.

3.3 Scaling with a Classifier

Hand-coding 87,000 sentences is not feasible, so we use the human labels to train a classifier. The model is a fine-tuned `bert-base-uncased` language model (Devlin et al., 2019).[1] The labeled sample has 2,732 sentences: 2,458 for training and 274 held out for testing. On the held-out set the model reached 90.1 percent accuracy and a macro-F1 score of 0.894.[2] These figures come from the original training run and are stored with the model. The labeled sentences themselves and the coders' individual labels were not preserved, so they cannot be re-estimated (Appendix B).

For each sentence the classifier gives a probability that it is restrictive, neutral, or supportive. We adjust these probabilities with temperature scaling before using them (Guo et al., 2017).[3]

3.4 The Index

The index places restrictive language at 0, neutral at 0.5, and supportive at 1. Each sentence receives the expected value of that scale under the classifier's probabilities. A sentence the model is confident is supportive scores near 1; one it is confident is restrictive scores near 0. Formally, if $\hat{p}_{Ri}$, $\hat{p}_{Ni}$, and $\hat{p}_{Si}$ are the adjusted probabilities for sentence i, its score is

$$s_i = 0.5\,\hat{p}_{Ni} + \hat{p}_{Si}. \tag{1}$$

PCSI for university j in year t is the simple average of its sentence scores:

$$\text{PCSI}_{jt} = \frac{1}{N_{jt}} \sum_{i=1}^{N_{jt}} s_i = 0.5 + 0.5\,(\bar{\hat{p}}_S - \bar{\hat{p}}_R)_{jt}\,. \tag{2}$$

The second form shows that a score below 0.5 means the policy carries more restrictive than supportive probability on average. It does not mean every sentence is restrictive; supportive and restrictive sentences can offset each other.

3.5 What the Index Captures

Three checks show what PCSI reflects.

First, the result does not depend on how sentence scores are combined. A trimmed mean, which drops the most extreme 5 percent of sentence scores, and a mean weighted by sentence length correlate with the baseline at 0.999 and 0.930. The median preserves most of the ranking of policies (rank correlation 0.878) but changes the distances between them more (Appendix C).

[1] Fine-tuning means training a model that has already learned general English further on the sentences labeled for this study. The model reads each word in the context of the words around it, which matters for terms such as *shall*, *may*, or *assign*.

[2] Accuracy is the share of held-out sentences given the same label as the human coders. Macro-F1 averages, across the three classes, a score that balances how often the model's predictions of a class are right and how many of that class's sentences it finds. It keeps the most common class from dominating the result.

[3] Temperature scaling rescales the probabilities with one fitted number so that they better reflect how often the model is right. It never changes which class is most likely.

Second, PCSI tracks the tone of the language rather than its readability or legal density. We build three simple word-based measures from the same text. *Tone* compares supportive and collaborative wording with obligation, control, and sanction wording; *Clarity* reflects sentence length and procedural guidance; *Legal Load* reflects legal and IP-technical vocabulary (Appendix E). Together they explain about a quarter of the variation in PCSI ($R^2 = 0.236$), and Tone accounts for almost all of that: 89 percent of the explained variation by a Shapley decomposition, against 9 percent for Legal Load and 2 percent for Clarity (Table 3).[4] Tone and Legal Load are nearly unrelated ($r = 0.029$): a technically dense policy can still offer help, and a plain one can assert firm control.

Table 3: Interpreting PCSI with Transparent Linguistic Features

	(1)	(2)	(3)
Tone Index	0.064***	0.063***	0.062***
	(0.012)	(0.012)	(0.012)
Clarity Index		0.001	0.006
		(0.008)	(0.008)
Legal Load Index			0.013**
			(0.006)
R^2	0.216	0.216	0.236
Shapley share of R^2: Tone / Clarity / Legal Load			0.89 / 0.02 / 0.09
Same, years with source text			0.78 / 0.14 / 0.09
Observations	4,277	4,277	4,277

Notes: Dependent variable is policy-in-force PCSI; 150 universities. Standard errors clustered by university are in parentheses. Shapley shares average each index's marginal contribution to R^2 over all orders of entry and sum to one. Each diagnostic combines standardized components, so one unit of a composite should not be read as one standard deviation of that composite. *** $p < 0.01$; ** $p < 0.05$; * $p < 0.10$.

Third, PCSI is not just a count of rules. Following the Institutional Grammar (Crawford and Ostrom, 1995), we use keyword rules to classify each sentence's deontic operator (the word that makes it an obligation, a prohibition, or a permission, such as *must*, *must not*, or *may*) and to flag sanction language. Obligations are the most restrictive form on average, with a mean sentence score of 0.329 against 0.467 for sentences without an operator. But most of the variation lies within forms: the eight combinations of operator and sanction explain only 4.8 percent of the sentence-level variation in stance (Appendix Table G3). A version of PCSI that removes these form differences correlates 0.94 with the original and gives the same main results (Appendix H). Keywords capture only part of a sentence's legal content, so this check shows that stance is not a count of musts and mays; it does not show that framing is fully separate from legal substance.

[4] A Shapley decomposition divides a regression's explained variation among its variables by averaging each variable's contribution over every possible order of entry. It describes fit, not causal importance (Grömping, 2006).

4 Results

4.1 Finding 1: University IP-Policy Language Leans Restrictive

Figure 1 shows the distribution of PCSI across the 4,277 annual observations, and Table 4 summarizes it. The mean is 0.437 and the median 0.427. Most observations lie between 0.39 and 0.47, and the full range runs from 0.264 to 0.730. The key fact is where this mass sits: 86.0 percent of observations fall below the neutral midpoint of 0.50. In the typical policy, the classifier assigns more weight to restrictive than to supportive language.

The pattern does not come from carrying old scores forward. Among the 480 university-years with new policy text, the mean is 0.429 and 89.6 percent fall below 0.50. The upper tail matters too. Some policies reach 0.73, so far more supportive writing is possible within the same genre.

Table 4: Distribution of PCSI

	Policy-in-force panel	Years with source text
Mean	0.437	0.429
Median	0.427	0.423
Standard deviation	0.066	0.063
25th percentile	0.394	0.391
75th percentile	0.471	0.464
Minimum	0.264	0.264
Maximum	0.730	0.730
Share below 0.50	86.0%	89.6%
Observations	4,277	480

Notes: The policy-in-force panel carries each university's most recent score forward until a newer policy source appears; the second column uses only university-years for which policy text is directly available. Analysis window 1944–2025; 150 universities.

4.2 Finding 2: Stance Has Become More Restrictive over Time

Figure 2 plots mean PCSI each year from 1978, the first year with at least five universities in the data. Average stance peaks at about 0.49 around 1980 and falls to about 0.42 by 2025.

Part of a raw trend like this can reflect changes in which universities the archive covers. The decline holds, however, when we follow the same universities. Among the 93 universities already in the data by 2000, mean PCSI falls from 0.447 to 0.422. A trend estimated within universities, which compares each university only with itself, is -0.00112 per year ($p = 0.008$).[5] Using only years with new policy text from 1980 onward gives almost the same estimate, -0.00117 per year ($p = 0.003$). Appendix F shows the same pattern for groups of universities fixed in 1990, 1995, and 2005.

[5]The within-university trends use university fixed effects, which remove each university's average level, and standard errors clustered by university, which allow for related observations from the same institution.

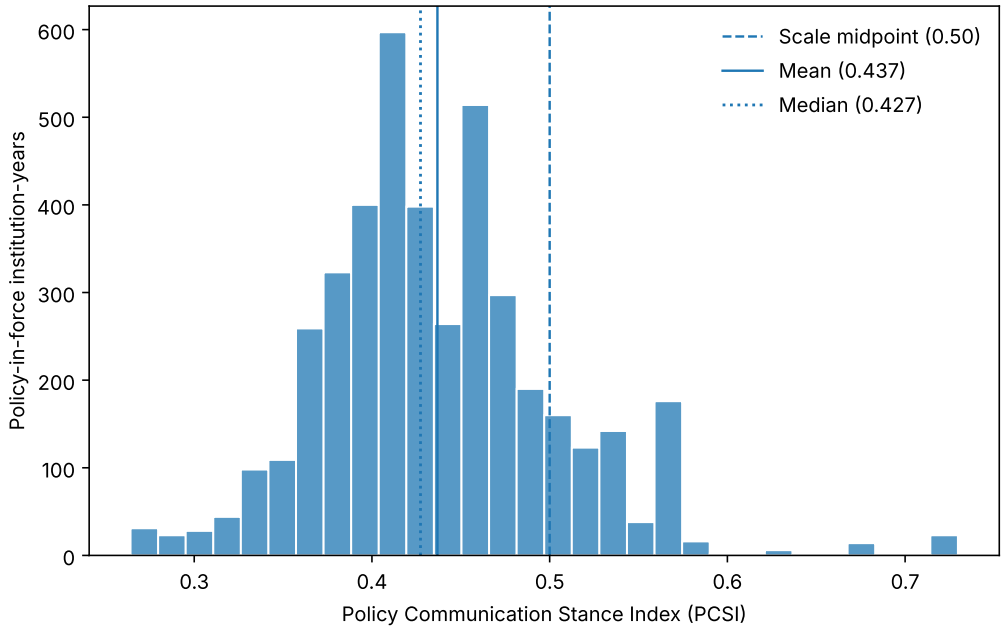


Figure 1: Distribution of PCSI in the 1944–2025 Policy-in-Force Panel ($N = 4{,}277$). The dashed line marks the scale midpoint of 0.50; the solid and dotted lines mark the mean and median.

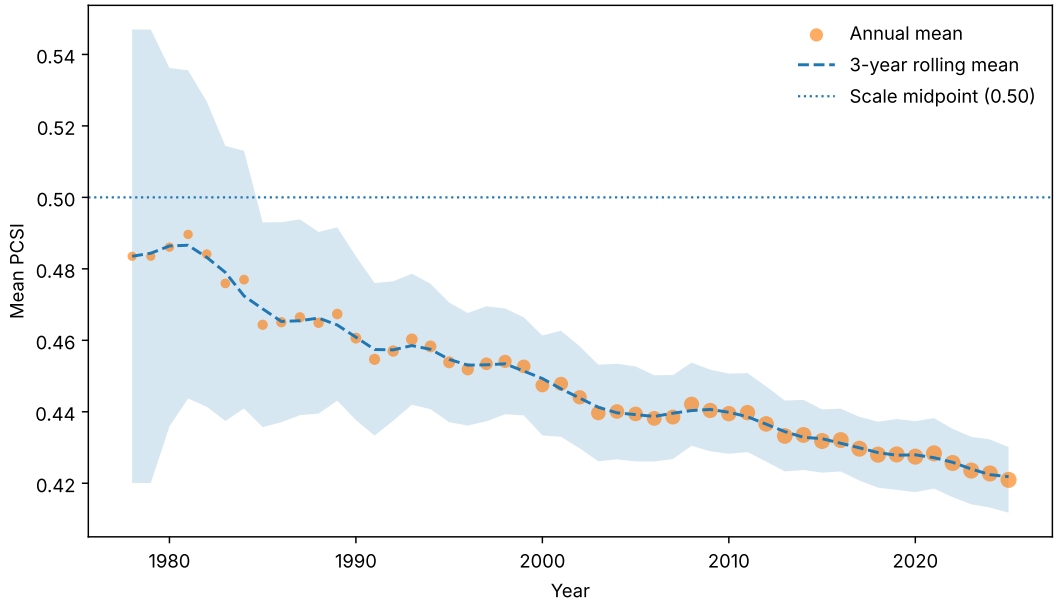


Figure 2: Annual Mean PCSI, 1978–2025. Points show each year's mean in the policy-in-force panel, with point size proportional to the number of universities. The shaded band is the 95% confidence interval, the dashed line is a three-year rolling mean, and the dotted line marks the midpoint of 0.50. Years before 1978 have fewer than five universities and are omitted.

A natural explanation is a change in the rules themselves. In *Stanford v. Roche* (2011), the Supreme Court distinguished a promise to assign patent rights from a present assignment of those rights.[6] Ownership and assignment clauses are also among the most restrictive in the corpus. If universities had simply rewritten these clauses, the decline would be concentrated there.

It is not. We sort sentences into clause topics by keyword and estimate the within-university trend for each (Figure 3). Ownership and assignment language shows almost no trend (-0.0003 per year, $p = 0.64$). Disclosure, revenue-sharing, and administrative language all become more restrictive, and every other topic moves in the same direction.

A second check splits the decline into two parts: changes in stance within categories of clauses, and changes in how much text falls in each category.[7] For the 63 universities with policy text in both 1980–1999 and 2010–2025, mean PCSI fell by 0.029. Changes within topics account for 77 percent of this decline and changes in the mix of topics for 23 percent. By rule form, changes within forms account for 94 percent. Within the keyword-defined categories, then, most of the decline comes from more restrictive language rather than from a shift toward more restrictive kinds of clauses. Because the categories are defined by keywords, this does not show that the underlying legal rules stayed the same; a change within a category could still reflect a change in legal content.

4.3 Finding 3: Universities Differ Persistently in Stance

Stance behaves like a stable trait of each university. In the annual panel, 61.0 percent of the variation in PCSI lies between universities and 39.0 percent within them over time. Carrying scores forward inflates this persistence, but using only years with new text still leaves 48.7 percent between universities (Appendix Table C2). Revisions tend to be small: across 330 successive revisions, the median change in PCSI is 0.021. The pattern resembles a lasting institutional writing style, much as individual managers leave distinctive styles on the firms they run (compare Bertrand and Schoar, 2003).

Some of these differences follow institutional type (Table 5). Private R1 universities average 0.456, against 0.427 for public R1 universities. Land-grant universities average 0.420, against 0.442 for other universities. Both differences are statistically significant in rank-based tests and survive a correction for testing several characteristics; medical-school status and urbanicity are not.[8]

Table 6 looks at these characteristics together. With one average per university, the private-R1 difference falls from 0.029 to about 0.020 once land-grant, medical-school, urbanicity, and

[6] *Board of Trustees of the Leland Stanford Junior University v. Roche Molecular Systems, Inc.*, 563 U.S. 776 (2011). The decision concerns assignment wording; by itself it does not show how many universities later revised their policies.

[7] This is a shift-share decomposition. We compute it separately for each university and then average, using the 63 universities with new policy text in both 1980–1999 and 2010–2025. Appendix G gives the exact formula.

[8] Rank-based tests (Kruskal–Wallis and Wilcoxon) compare groups using the ranks of the values, which suits small groups and skewed data. The Holm correction adjusts the four p-values for multiple testing (Appendix Table D1). We label universities without the data's R1 flag "non-R1"; the flag's Carnegie edition is not recorded, so we do not call this group R2. Land-grant status follows the Morrill Acts of 1862 and 1890, and urbanicity is a three-way rural, semi-urban, and city coding supplied with the data.

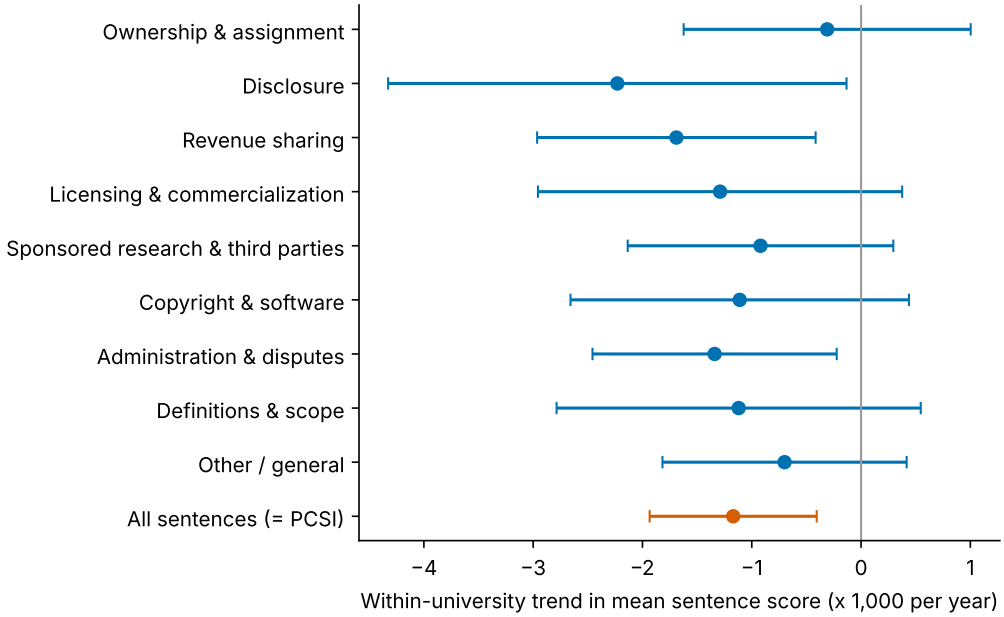


Figure 3: Within-University Trends in Sentence Stance by Clause Topic, 1980–2025. Each point is the yearly trend in a topic's mean sentence score within universities, using years with new policy text from 1980 onward; trends are multiplied by 1,000. Bars are 95% confidence intervals with standard errors clustered by university. Topics are assigned by keyword and can overlap (Appendix G).

Table 5: University-Level PCSI by Institutional Characteristics

Group	*N*	Mean	SD	Range
Carnegie classification and control				
Private R1	38	0.456	0.046	[0.337, 0.534]
Public R1	86	0.427	0.051	[0.328, 0.585]
Private non-R1	8	0.406	0.053	[0.342, 0.459]
Public non-R1	18	0.441	0.058	[0.376, 0.580]
Mission and program				
Land-grant	47	0.420	0.047	[0.338, 0.585]
Non-land-grant	103	0.442	0.053	[0.328, 0.580]
Medical school	118	0.438	0.053	[0.328, 0.585]
No medical school	32	0.422	0.047	[0.342, 0.519]
Urbanicity				
Rural	15	0.425	0.041	[0.372, 0.519]
Semi-urban	48	0.422	0.047	[0.338, 0.539]
City	82	0.443	0.057	[0.328, 0.585]

Notes: Unit of analysis is the university; PCSI is averaged across each university's policy-in-force years before group statistics are computed. Five universities lack urbanicity data. Rank-based tests are reported in Appendix D.

region are included, and it is no longer precisely estimated. University averages, however, mix institutional differences with the decades in which each policy was written. Column (4) therefore compares only policies written in the same year. Here the private-R1 difference is 0.040 (standard error 0.012), while the land-grant difference shrinks to −0.005 and is imprecise. The private-R1 gap is the clearest institutional pattern; the land-grant gap largely reflects overlap with other characteristics. Weighting universities by the number of documents or sentences does not change the overall picture (Appendix Table D2).

Table 6: Institutional Correlates of PCSI

	University means			Source-text years
	(1)	(2)	(3)	(4)
Private R1	0.029***	0.019	0.021	0.040***
	(0.009)	(0.012)	(0.014)	(0.012)
Private non-R1	−0.021	−0.031	−0.035	−0.006
	(0.021)	(0.023)	(0.023)	(0.018)
Public non-R1	0.014	0.007	0.011	0.005
	(0.015)	(0.020)	(0.018)	(0.011)
Land-grant		−0.014	−0.011	−0.005
		(0.012)	(0.013)	(0.012)
Medical school		0.014	0.010	0.006
		(0.011)	(0.011)	(0.009)
Urbanicity controls	No	Yes	Yes	Yes
Region fixed effects	No	No	Yes	Yes
Source-year fixed effects	No	No	No	Yes
R^2	0.074	0.111	0.161	0.291
Observations	150	145	144	452
Universities	150	145	144	144

Notes: Dependent variable is PCSI. Columns (1)–(3) use one observation per university (mean policy-in-force PCSI) with HC3 robust standard errors; Columns (2)–(3) omit universities with missing covariates. Column (4) uses university-years with directly available policy text, compares policies written in the same year through source-year fixed effects, and clusters standard errors by university. Public R1, rural, and Midwest are the omitted categories. *** $p < 0.01$; ** $p < 0.05$; * $p < 0.10$.

Even taken together, observable characteristics explain at most about a sixth of the variation in average PCSI across universities. Broad categories organize some of the differences in how universities write, but most of the variation is specific to each institution.

5 Discussion

The three findings fit together. University IP policies tend to address inventors in restrictive terms; that tendency has grown since the 1980s across many kinds of clauses; and each university keeps a recognizable stance that broad institutional categories explain only in part. The first finding

shows that stance is a real feature of these documents. The second and third show that it varies in the ways the literature on organizational texts and institutional fields would lead one to expect.

Persistence with weak observable correlates is consistent with drafting histories specific to each institution. Research on contracts shows that legal language is often carried over from earlier documents and changed in small steps (Kahan and Klausner, 1997; Choi et al., 2013); university policies may evolve in a similar way, although those studies do not examine universities. The shared shift toward more restrictive wording fits the view that organizations in the same field respond to common legal and professional pressures (DiMaggio and Powell, 1983; Edelman, 1992). Because the shift appears within clause topics and rule forms, it looks more like a change in drafting register than a response to one legal event.

Why might written stance matter? The literature reviewed in Section 2 points to the people who administer the rules. TTO staff and administrators interpret policies, run disclosure procedures, and negotiate with inventors, and faculty decisions respond to how costly and supportive those interactions seem (Owen-Smith and Powell, 2001; Jensen et al., 2003; Siegel et al., 2003; Bercovitz and Feldman, 2008). A policy's stance may shape the guidance and expectations these intermediaries work from. This is a possible pathway, not one we test. PCSI measures the written policy, not how TTO staff behave, how faculty perceive governance, or what they go on to do.

The study has several limitations. The annual panel assumes that the most recent policy stays in force until a newer one is found; results using only years with new text are similar, but the archive may be incomplete. The classifier reads sentences one at a time, so it misses layout and the links between passages, and the keyword rules for topics and rule forms are transparent but coarse. Our regressions treat predicted scores as data. The main results hold when PCSI is built from each sentence's single most likely label, or only from sentences the model classifies with high confidence (Appendix H), but formal corrections for predicted variables are not applied (Egami et al., 2023; Battaglia et al., 2024). Finally, a more supportive policy is not necessarily a better one, and PCSI says nothing about the generosity of legal rights.

6 Conclusion

University IP policies are usually studied for what they require. This paper measures how they say it. PCSI anchors supportive and restrictive language in human judgment and scales it to 150 research universities over eight decades. As scored by the classifier, the policies lean restrictive, have become more restrictive since the 1980s, and differ persistently across universities in ways that observable characteristics explain only partly. The measure opens a path for studying how written governance relates to the practices of technology-transfer offices and to the decisions of the faculty they serve.

Data and Code Availability

The scored sentences, derived indices, code, and manuscript source are available at https://github.com/krish533/PCSI.

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A Data Construction

The analysis uses 87,159 cleaned sentences from 518 policy documents at 150 U.S. research universities, 1944–2025. Documents from the same university and year are pooled before scoring, giving 480 university-years with policy text. The annual panel carries each score forward until a newer document appears for that university, giving 4,277 observations through 2025; each row records the year of the document behind its score. For comparisons across universities, each university's annual scores are averaged so that every university counts once. The archive also holds a single one-sentence record from 1925, which is kept for completeness and excluded from the analysis.

B Coding Rubric and Classifier

Coding rubric. Coders label each sentence supportive, neutral, or restrictive according to its main message. Supportive language emphasizes help, guidance, encouragement, collaboration, shared interests, or facilitation. Restrictive language emphasizes mandatory assignment, unilateral authority, prohibitions, sanctions, forfeiture, or categorical compliance. Neutral language mainly defines terms, describes arrangements, or sets out procedure. Several trained coders applied the rubric, and disagreements were adjudicated.

What is preserved. The released package contains the trained classifier, its configuration and tokenizer, the calibration parameter, and the metrics recorded when the model was trained (2,458 training and 274 validation sentences; 90.1 percent accuracy; macro-F1 of 0.894). The 2,732 adjudicated sentences and the coders' individual labels are not preserved. The accuracy figures are therefore historical results stored with the model, and agreement among the original coders cannot be reported.

Classifier and calibration. Table B1 reports the model's settings. Temperature scaling is applied before probabilities are averaged. Rescoring a random sample of 2,000 sentences with the saved model reproduces the stored probabilities to within 4×10^{-6} and gives every sentence the same label. Rebuilding the university-year indices from the stored sentence scores reproduces them exactly.

C Aggregation and Persistence

Table C1 compares the baseline index with the median, a 5-percent trimmed mean, and a length-weighted mean. Midpoint agreement is the share of observations that both versions place on the same side of 0.50. Trimming changes almost nothing, and length weighting stays close to the baseline. The median changes levels the most but keeps a rank correlation of 0.878. Table C2 splits

Table B1: Preserved Classifier Configuration

Parameter	Value
Base model	`bert-base-uncased`
Transformer layers	12
Attention heads	12
Hidden size	768
Intermediate size	3,072
Maximum input length	256 tokens
Output classes	3
Training sentences	2,458
Validation sentences	274
Total labeled sentences	2,732
Temperature parameter	$T = 0.8462$
Validation accuracy	0.901
Validation macro-F1	0.894

Notes: Configuration and metrics are read from the preserved model artifacts in the replication package.

the variation into between- and within-university parts for each version. Carrying scores forward raises persistence in every version, but large differences across universities remain without it.

Table C1: Sensitivity to Alternative Sentence Aggregation Rules

Aggregation	Mean	Pearson r	Spearman ρ	Midpoint agreement
Baseline (unweighted mean)	0.437	1.000	1.000	1.000
Median PCSI	0.471	0.623	0.878	0.937
5% trimmed mean	0.431	0.999	1.000	1.000
Length-weighted mean	0.433	0.930	0.906	0.933

Notes: $N = 4{,}277$ policy-in-force university-years. Correlations are with baseline PCSI. Midpoint agreement is the share of observations that the baseline and the alternative place on the same side of 0.50.

D Institutional Comparisons

Table D1 reports the four rank-based comparisons with Holm-adjusted p-values. The simple differences hold across ways of combining sentence scores: the private-R1 minus public-R1 gap is 0.029 with the baseline, 0.018 with the median, and 0.032 with the trimmed and length-weighted means, and the land-grant gap ranges from -0.020 to -0.024. Using only years with new text, the two gaps are 0.033 and -0.019.

Table D2 reports the private-R1 and land-grant coefficients across samples and weights. The weighted rows give more influence to universities with more documents or sentences. Archive timing differs a little by type: the average document year is 2002.3 for private R1 universities and 2005.1 for public R1 universities. Because PCSI falls over time, university averages slightly favor

Table C2: Between- and Within-University Variation

Specification	Between share	Within share
Policy-in-force panel		
Baseline	0.610	0.390
Median	0.434	0.566
Trimmed mean	0.616	0.384
Length-weighted mean	0.589	0.411
Years with source text		
Baseline	0.487	0.513
Median	0.300	0.700
Trimmed mean	0.494	0.506
Length-weighted mean	0.463	0.537

Notes: Shares partition the total sum of squared deviations of PCSI into between- and within-university components.

Table D1: Institutional Comparisons with Multiple-Testing Adjustment

Characteristic	Test	Raw p	Holm p
Carnegie classification and control	Kruskal–Wallis	0.004	0.014
Land-grant status	Wilcoxon rank-sum	0.010	0.029
Urbanicity	Kruskal–Wallis	0.055	0.111
Medical-school status	Wilcoxon rank-sum	0.172	0.172

Notes: University-level mean PCSI ($N = 150$; 145 for urbanicity). The Holm adjustment is applied to this family of four tests.

older archives, which is why Column (4) of Table 6 compares policies written in the same year. The private-R1 gap is larger in that comparison, not smaller.

Table D2: Private-R1 and Land-Grant Coefficients Across Samples and Estimators

	Type only		Full controls		
	Private R1	N	Private R1	Land-grant	N
University means	0.029***	150	0.021	−0.011	144
	(0.009)		(0.014)	(0.013)	
WLS, weight = source policy years	0.027**	144	0.020	−0.021	144
	(0.010)		(0.015)	(0.015)	
WLS, weight = sentences	0.021	144	0.023	0.005	144
	(0.014)		(0.023)	(0.028)	
Era-adjusted university means	0.029***	144	0.022	−0.009	144
	(0.009)		(0.014)	(0.013)	
Source-text years, year FE	0.040***	452	0.040***	−0.005	452
	(0.010)		(0.012)	(0.012)	
Source-text years from 1980, year FE	0.039***	439	0.038***	−0.005	439
	(0.010)		(0.012)	(0.012)	

Notes: Dependent variable is PCSI; public R1 is the omitted type. "Full controls" adds land-grant, medical-school, and urbanicity indicators and region fixed effects; it uses the 144 universities with complete covariates. Re-estimating the type-only specification on those same 144 universities gives a private-R1 coefficient of 0.029*** (0.009), so the change between the two columns is not due to the smaller sample. University-mean rows use HC3 standard errors. Era-adjusted means subtract each calendar year's cross-sectional mean from the policy-in-force panel before averaging within university. Source-text rows use university-years with directly available policy text, include source-year fixed effects, and cluster standard errors by university. *** $p < 0.01$; ** $p < 0.05$; * $p < 0.10$.

E Word-Based Diagnostics

The three diagnostics help interpret PCSI; they are not alternative measures of stance. Each component is standardized (expressed relative to its mean and standard deviation) before the components are combined:

$$\begin{aligned}\text{Tone} = \tfrac{1}{6}\big[&z(\text{supportive}) + z(\text{second person}) + z(\text{inclusive we}) \\ &- z(\text{restrictive}) - z(\text{sanction}) - z(\text{obligation modal share})\big],\end{aligned} \tag{3}$$

$$\text{Clarity} = \tfrac{1}{3}\big[-z(\text{mean sentence length}) - z(\text{long-sentence share}) + z(\text{procedural share})\big], \tag{4}$$

$$\text{Legal Load} = \tfrac{1}{2}\big[z(\text{legalese}) + z(\text{IP technical})\big]. \tag{5}$$

The supportive, restrictive, sanction, procedural, legalese, and IP-technical components come from word lists in the replication code, and the second-person and inclusive-"we" components capture direct and collective forms of address. The diagnostics are distinct: Tone and Clarity correlate at 0.143, Tone and Legal Load at 0.029, and Clarity and Legal Load at −0.296. Table E1 shows that

controlling for policy length leaves the Tone relationship almost unchanged. In the 480 years with new text, the three-diagnostic regression gives coefficients of 0.038 for Tone, 0.013 for Clarity, and 0.012 for Legal Load, with $R^2 = 0.151$.

Table E1: Linguistic Decomposition with Policy-Length Controls

	Baseline	+ Log words	+ Log sentences
Tone Index	0.062***	0.059***	0.059***
	(0.012)	(0.012)	(0.012)
Clarity Index	0.006	0.006	0.008
	(0.008)	(0.008)	(0.009)
Legal Load Index	0.013**	0.010	0.010
	(0.006)	(0.007)	(0.007)
Policy-length control	No	Log words	Log sentences
R^2	0.236	0.244	0.241
Observations	4,277	4,277	4,277

Notes: University-clustered standard errors in parentheses. *** $p < 0.01$; ** $p < 0.05$; * $p < 0.10$.

F Trends over Time

Table F1 follows fixed groups of universities from several start years. Every group has a lower mean in 2025 than at its start, and every within-university trend is negative. Table F2 uses only years with new policy text and reports both a pooled trend and a within-university trend. The post-2000 estimate is negative but less precise. The 2025 cross-section is not driven by old policies: among the 70 universities whose 2025 policy is at most five years old, 91 percent fall below the midpoint, against 92 percent of all 150.

Table F1: PCSI Trends Within Fixed University Cohorts

Cohort start	Universities	Start mean	2025 mean	FE trend per year	p
1990	34	0.461	0.421	−0.00109	0.019
1995	53	0.454	0.422	−0.00131	0.002
2000	93	0.447	0.422	−0.00112	0.008
2005	114	0.439	0.420	−0.00103	0.011

Notes: Each cohort contains the universities already represented in the policy-in-force panel by the start year. Trends come from regressions of PCSI on year with university fixed effects; standard errors are clustered by university.

G Clause Topics and Rule Forms

Clause topics. Sentences are sorted into topics by keyword: ownership and assignment (*assign*, *ownership*, *title*, *vest*, *belong*); disclosure (*disclos*, *invention report*); revenue sharing (*royalt*, *revenue*,

Table F2: Temporal Trends Using Only Years with Source Policy Text

Start year	N	Pooled trend	FE trend	SE	p
1980	467	−0.00115	−0.00117	(0.00039)	0.003
1990	443	−0.00121	−0.00128	(0.00053)	0.016
2000	367	−0.00109	−0.00150	(0.00093)	0.105

Notes: University-years with directly available policy text from the start year onward. Standard errors are clustered by university.

net income/proceeds, *distribut*, *equity*); licensing and commercialization (*licens*, *commerciali*, *technology transfer*, *start-up*, *patent application*); sponsored research and third parties (*sponsor*, *grant*, *federal*, *government*, *contract*, *third part*); copyright and software (*copyright*, *software*, *courseware*, *scholarly work*); administration and disputes (*appeal*, *dispute*, *committee*, *review*, *waive*, *grievance*, *amend*); and definitions and scope (*means*, *defined*, *applies to*, *purpose*, *scope*). In the topic trends a sentence can belong to several topics. In the decomposition each sentence gets one main topic (the first match in this order), and unmatched sentences form an "other" category. Among sentences from 1980 onward, 59 percent match at least one topic. Ownership and assignment sentences are the most restrictive (mean score 0.335) and definition and revenue-sharing sentences the least (0.448 and 0.453). Table G2 reports the topic trends.

Decomposition. The decomposition compares 1980–1999 with 2010–2025 for the 63 universities with new text in both periods (8,353 and 28,175 sentences). For university i in period $t \in \{0, 1\}$, let s_{ikt} be the share of its sentences in category k and m_{ikt} the category's mean score, so the university's mean is $M_{it} = \sum_k s_{ikt} m_{ikt}$. Then

$$M_{i1} - M_{i0} = \underbrace{\sum_k \frac{s_{ik1} + s_{ik0}}{2} (m_{ik1} - m_{ik0})}_{\text{within categories}} + \underbrace{\sum_k \frac{m_{ik1} + m_{ik0}}{2} (s_{ik1} - s_{ik0})}_{\text{composition}}. \tag{6}$$

If a category is missing for a university in one period, its mean there is set to its mean in the other period, which assigns the category's appearance or disappearance to composition. The identity is applied university by university and the two parts are then averaged with equal weights. Both decompositions add up exactly to the observed change. The topic and rule-form versions are alternative descriptions of the same change and should not be added together. Table G1 reports the results.

Rule forms. Each sentence gets one deontic operator, in this order of priority: prohibition (a negated *shall*, *must*, *may*, *will*, or *can*; *cannot*; *prohibit*; *not permitted*), obligation (*shall*, *must*, *is required*, *requires*, *obligat-*, *agrees to*, *is expected to*), permission (*may*, *permit*, *entitled*, *discretion*, *can*, *allow*), or none. Sanction language (*violat-*, *disciplin-*, *sanction*, *forfeit*, *terminat-*, *breach*, *penalt-*, *dismiss*, *revok-*) is flagged separately. Table G3 reports stance by form. At the policy level, the

Table G1: Exact Shift–Share Decompositions of the Modern PCSI Decline

	Exclusive primary topic	Deontic × sanction
Early period: mean score (1980–1999)	0.455454	0.455454
Late period: mean score (2010–2025)	0.426638	0.426638
Observed change	−0.028816	−0.028816
Within-category contribution	−0.022229	−0.027077
Composition contribution	−0.006587	−0.001739
Within-category share of decline	77.1%	94.0%
Composition share of decline	22.9%	6.0%
Universities with policies in both periods	63	63
Early and late sentences	8,353; 28,175	8,353; 28,175

Notes: The unit of comparison is a university appearing in both periods; universities are weighted equally. Each sentence is assigned to one primary topic by fixed keyword priority, or to one of eight deontic-by-sanction cells. Category changes are decomposed within each university using midpoint shares and midpoint conditional means, then averaged. If a category is absent in a period, its conditional mean is imputed from the observed period, assigning category entry/exit to composition. The university-level accounting identity holds to machine precision. These alternative decompositions are not additive and are sensitive to the coding rules and missing-category convention. The numbers are descriptive, not causal estimates.

shares of obligation, prohibition, permission, and sanction sentences explain 11 percent of the variation in PCSI across the 480 years with new text.

H Alternative Versions of PCSI

PCSI averages probabilities rather than single labels, so it keeps track of the classifier's uncertainty, but the regressions still treat predicted scores as data. Table H1 recomputes the main results three ways: from each sentence's single most likely label (scored 0, 0.5, or 1); from the 66 percent of sentences whose most likely label has a probability of at least 0.9; and from the version that removes rule-form differences (Section 3.5). Every version places most policies below the midpoint, attributes about half of the variation in years with new text to universities, shows a negative within-university trend, and gives a positive and significant private-R1 coefficient. The model's confidence is also stable over time: from the 1970s onward, the average probability of the most likely label lies between 0.874 and 0.886 in every decade.

Table G2: Within-University Trends in Sentence Stance by Clause Topic, 1980–2025

Topic	Share	Mean score	FE trend per year	SE	p
Ownership & assignment	0.142	0.335	−0.00031	(0.00067)	0.644
Disclosure	0.057	0.357	−0.00223	(0.00107)	0.037
Revenue sharing	0.131	0.453	−0.00169	(0.00065)	0.010
Licensing & commercialization	0.133	0.432	−0.00129	(0.00085)	0.129
Sponsored research & third parties	0.131	0.341	−0.00092	(0.00062)	0.140
Copyright & software	0.092	0.412	−0.00111	(0.00079)	0.159
Administration & disputes	0.141	0.425	−0.00134	(0.00057)	0.018
Definitions & scope	0.083	0.448	−0.00112	(0.00085)	0.185
Other / general	0.409	0.427	−0.00070	(0.00057)	0.213
All sentences (= PCSI)	1.000		−0.00117	(0.00039)	0.002

Notes: Each row regresses the university-year mean sentence score within a topic on year and university fixed effects, using source-text university-years from 1980 and universities observed in at least two such years; standard errors are clustered by university. Topics are assigned by keyword rules and can overlap; share is the fraction of 1980–2025 sentences matching the topic. Within-topic trends allow overlapping keyword matches. The exact shift-share calculation instead uses mutually exclusive primary topics and is reported separately in Table G1.

Table G3: Communicative Stance by Rule Form

Rule form	Share	Mean score	Restrictive	Supportive
Obligation	0.314	0.329	0.466	0.105
Prohibition	0.072	0.430	0.470	0.316
Permission	0.138	0.431	0.406	0.246
No deontic operator	0.475	0.467	0.226	0.160
Sanction language (any form)	0.037	0.334	0.487	0.124

Notes: Deontic operators follow the Institutional Grammar of Crawford and Ostrom (1995) and are assigned by keyword rules with priority prohibition > obligation > permission (Appendix G). Sanction language (the grammar's "or else" component) can co-occur with any form. Restrictive and supportive shares use the classifier's most likely label. Rule form explains 4.8 percent of sentence-level variance in stance. The exact 63-university within-cell and composition changes are reported in Table G1.

Table H1: Headline Results Under Alternative Constructions of PCSI

Construction	r with baseline	Mean	Share < 0.50	Between share	FE trend, 1980+	Private R1
Baseline PCSI		0.429	0.896	0.487	−0.00117***	0.040***
					(0.00039)	(0.012)
Hard labels	0.980	0.425	0.856	0.487	−0.00139***	0.042***
					(0.00044)	(0.013)
High-confidence sentences	0.952	0.418	0.875	0.495	−0.00134***	0.049***
					(0.00051)	(0.014)
Framing net of rule form	0.943	0.427	0.894	0.492	−0.00135***	0.032**
					(0.00037)	(0.013)

Notes: All columns use the 480 university-years with directly available policy text. Hard labels score each sentence 0, 0.5, or 1 by its most likely class. High-confidence uses only sentences whose largest calibrated class probability is at least 0.9 (about two-thirds of sentences). Framing net of rule form subtracts the mean score of each sentence's deontic-by-sanction cell before aggregation. The trend is the coefficient on year with university fixed effects, using universities with at least two source-text years from 1980; the private-R1 coefficient is from the Column (4) specification of Table 6. Standard errors clustered by university. *** $p < 0.01$; ** $p < 0.05$; * $p < 0.10$.